

Reduction of gas emission via optimization of purified purge gas recycle ratio for conversion of CO₂ to methanol



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ABSTRACT

The current study highlights the significance of optimum recycle purge gas ratio (Φ) to methanol synthesis reactor calculated via a one dimensional modeling of process and also proposes an environmentally friendly process. Moreover, Differential Evolution (DE) method is applied to maximize methanol mole fraction. The purge gas of this process includes about 60% hydrogen, 20% carbon monoxide and carbon dioxide and 20% nitrogen and methane. Burning of such these gases is not economical and environmentally friendly process, thus recycling the purge gas is a reliable work. Although complete restoration of the purge gas is environmentally friendly, it is not an economical process. In order to prevent accumulation of inert gases such as nitrogen and methane, a purge gas line is attached to the recycling stream. In addition, a nitrogen purification unit has been proposed in the present study. Nitrogen is separated from the recycle gas stream by a novel metallic membrane; furthermore, nitrogen is converted to ammonia during the purification process. The increase in methanol production in Optimized Reactor Configuration (ORC) is 41% compared with the one in Industrial Configuration (IC). The optimum recycle ratio is 0.41 in ORC while this parameter is 2.19 in IC and zero in Fresh Feed Configuration (FFC). Simulation results show that Stoichiometry Number (SN) is reduced to 2.1 at the entrance of reactor; moreover, methanol production rate increases in ORC. Furthermore, the capability of ORC for producing ammonia is 305 mol/h. Enhancement in methanol and ammonia productions demonstrate the superiority of ORC to the previous configurations.

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1. Introduction

It has been argued that nowadays environmental and energy challenges facing humanity are becoming two important and crucial factors which have limited social and economical development. Application of new environmentally friendly and highly economical approaches is essential in order to satisfy the ever-increasing global energy demand [1]. Purge gas emission of methanol synthesis plant, which contains hazardous components, is regarded as one of the major sources of this study. Furthermore, valuable components such as hydrogen are included in such these purge gas streams. During methanol production, a portion of un-reacted syngas and some inert gases must be emitted to increase syngas rate of methanol production unit.

As a matter of fact, a large amount of natural gas, which could be utilized as a source of energy, is flared in industrial plants annually. This amount is approximately equal to 150 billion cubic meters,

which is almost equivalent to about 30% of EU natural gas use, 23% of US natural gas consumption and almost 5% of global natural gas production [2].

It is obvious that a wide range of potential fuels, which could be utilized to provide energy, are burnt. Releasing plenty of carbon dioxide gas in the atmosphere has been enhanced by burning flare gases, which has a great impact on both climate and environment. By burning these gases the opportunity to reuse them for various purposes, especially in areas which have difficulties in the energy supply, has been lost.

Enormous quantities of co-produced gases are flared as a waste by-product and large supplies of gas have emerged in order to fulfill the ever-increasing global demand for oil and gas. Therefore, the unwanted components can be removed and valuable components can be recycled to the feed stream. Not only a considerable amount of valuable energy wastes, but also severe environmental problems in the petroleum and related industries have been contributed to the source of much controversial debate [3].

Burning off any unwanted gas that liberates from oil during production is a common practice among all oil producing companies. The implementation of no-flare design will have a great impact on

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Nomenclature

a_v	Specific surface area of catalyst pellet ($\text{m}^2 \text{m}^{-3}$)
A_c	Cross section area of each tube (m^2)
c	Total concentration (mol m^{-3})
C_p	Specific heat of the gas at constant pressure ($\text{J mol}^{-1} \text{K}^{-1}$)
d_p	Particle diameter (m)
D	Reactor diameter (m)
F	Total molar flow rate (mol s^{-1})
h	Gas–solid heat transfer coefficient ($\text{W m}^{-2} \text{K}^{-1}$)
ΔH_i	Enthalpy of reaction
$\Delta H_{f,i}$	Enthalpy of formation of component i (J mol^{-1})
k_g	Mass transfer coefficient for component i (m s^{-1})
K	Conductivity of gas phase ($\text{W m}^{-1} \text{K}^{-1}$)
k_i	Reaction rate coefficient ($\text{kmol kgcat}^{-1} \text{h}^{-1}$)
K_i	Adsorption equilibrium constant (bar^{-1})
K_{WGS}	Equilibrium constant for water–gas shift reaction
P	Total pressure (for exothermic side: bar; for endothermic side: Pa)
r_i	Reaction rate of component i ($\text{mol kg}^{-1} \text{s}^{-1}$)
R	Universal gas constant ($\text{J mol}^{-1} \text{K}^{-1}$)
T	Temperature (K)
u_g	Gas velocity (m s^{-1})
U	Overall heat transfer coefficient between exothermic and endothermic sides ($\text{W m}^{-2} \text{K}^{-1}$)
y_i	Mole fraction of component i (mol mol^{-1})
z	Axial reactor coordinate (m)

Greek letters

ε_s	Void fraction of catalyst
η	Catalyst effectiveness factor
μ	Viscosity of fluid phase ($\text{kg m}^{-1} \text{s}^{-1}$)
ρ	Density of fluid phase (kg m^{-3})

Superscripts

b	Bed
g	In the bulk gas phase
s	At the surface of the catalyst

Subscripts

0	Inlet conditions
i	Chemical species in solid phase
j	Chemical species in fluid phase
k	Reaction number index

decreasing the emissions from production. It is expected that gas flaring will not be allowed in the near future with increasing awareness of the environmental impact and the ratification of the Kyoto protocol by most of the member countries. This will require significant changes in the current practices of oil and gas production and processing [4]. The results of two case studies of reducing, recovering and reusing flare gases from the Tabriz Petroleum Refinery and Shahid Hashemi-Nejad (Khangiran) Natural Gas Refinery, both in Iran is presented by Zadakbar et al. [5]. Determination of greenhouse gas emission resulting from gas flaring activities in Nigeria is investigated by Anomohanran et al. [6]. In the commercial processes, the unconverted syngas is recycled back to the methanol converter for enhancing the overall conversion, thereby improving the process economics. Purge gas of ammonia synthesis plant which contains hazardous components is recovered by Siavashi et al. [7]. The integrated configuration of catalytic

hydrogen-perm-selective membrane reactor and solid oxide fuel cell as a novel technology are applied in their study.

There have been numerous studies conducted to develop the efficiency of the industrial methanol synthesis reactor [8,9].

Some species such as nitrogen and methane are not beneficial for methanol synthesis. Therefore, a nitrogen and methane separation is effective in this study. Nitrogen can be separated by membrane. It follows that the removed nitrogen can be converted to ammonia.

1.1. Nitrogen removal

Metallic membranes, i.e., vanadium (V), made of Earth-abundant group V metals, are applied in order to separate nitrogen for ammonia synthesis with indirect carbon dioxide capture. This membrane, which selects nitrogen, profits from nitrogen driving force in flue gas streams (73 wt%) for the purpose of carbon dioxide capture. Consequently, it can create an adequate external driving force, for example, partial pressure across a membrane. Nitrogen selective membrane concept is due to development of metallic membranes which select hydrogen for hydrogen purification processes. There are a vast variety of applications for hydrogen selective metallic membrane in various fields. It is due that they may be fabricated into large-scale continuous films and also their commercial availability [10]. Group V metals are located at the far left of the transition metals in the periodic table. They are known to bind adsorbates very strongly to serve as effective catalysts. However, according to this property, these metals may act to dissociate successfully and transport atomic nitrogen through their crystal lattice. A schematic of nitrogen-selective membrane reactor for a post-combustion capture application is illustrated in Fig. 1. Nitrogen molecules are preferentially adsorbed on the catalytic membrane and dissociated into two nitrogen atoms. This phenomenon similarly occurs in a traditional palladium-based hydrogen selective membrane.

The atomic nitrogen permeates through the crystal lattice by hopping through the interstitial crystal sites of the bulk metal, ultimately leading to atomic nitrogen on permeable (or sweep) side of membrane [11–13].

Fick's first law of diffusion and Sieverts' law are utilized to calculate nitrogen flux in metals. In 1996, Buxbaum et al. [14] described an equation for hydrogen transportation in metals, which was well adapted to the survey in this study. The nitrogen flux equation is as follows:

$$J_{\text{N}_2} = \frac{P_{\text{M}}}{\delta} (\sqrt{P_{\text{h}} x \text{N}_2} - \sqrt{P_{\text{j}} y \text{N}_2}) \quad (1)$$

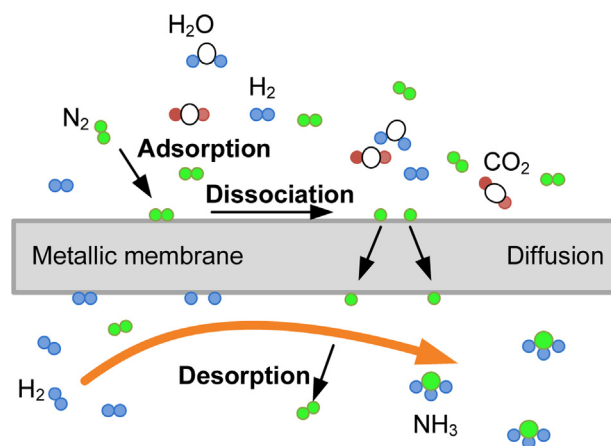


Fig. 1. Schematic diagram of a nitrogen-selective membrane reactor.

where J_{N_2} is nitrogen flux [$\text{mol}/\text{m}^2 \text{ s}$], P_M is permeability of metal [$\text{mol}/\text{m s Pa}^{1/2}$], and δ is membrane thickness [m]. The ratio of P_M/δ is defined as the permeance. P_h and P_l are the total gas pressures [Pa] on the feed and permeate sides, respectively, and X_{N_2} and Y_{N_2} are the nitrogen mole fractions of the feed-side and permeate-side, respectively.

1.2. Objective

In this study, optimization has been performed and the optimum recycle ratio of purge gas has been obtained. Recycle ratio (Φ) is defined as:

$$\Phi = \frac{\text{recycle}}{\text{MUG}} \quad (2)$$

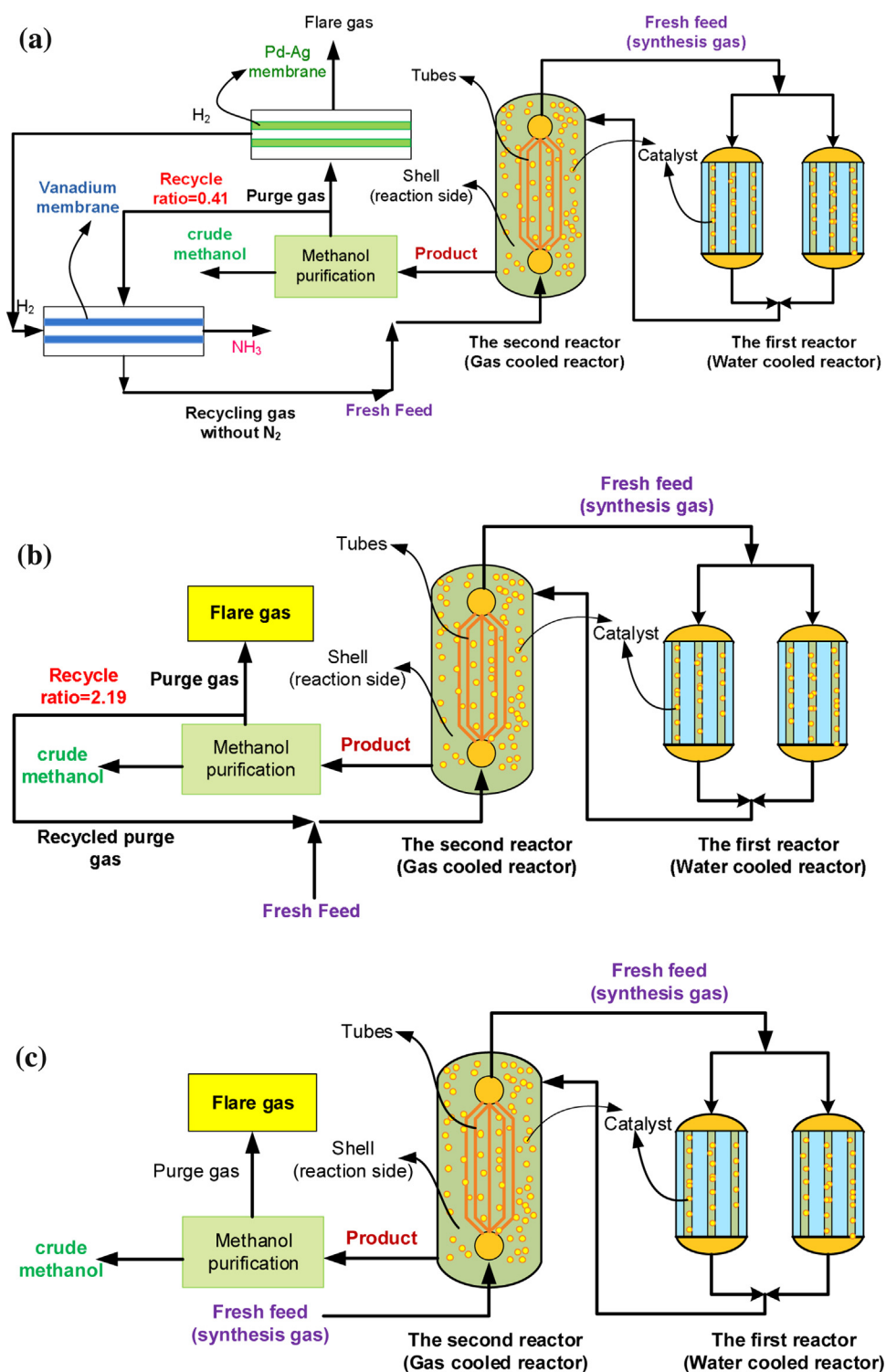


Fig. 2. Schematic diagram of different configurations (a) ORC, (b) IC and (c) FFC.

where MUG is makeup gas [15].

The maximum methanol production occurs in optimum recycle ratio of purge gas which is simulated in the proposed configuration named as Optimized Recycle Configuration (ORC). In this configuration, two purification units are essential. One of them is nitrogen removal from purge gas in order to be recycled to the methanol synthesis reactors. Another one is hydrogen separation from the flare gas to ammonia production in the membrane reactor. The results then have been compared with Fresh Feed Configuration (FFC) and Industrial Configuration (IC). There are three advantages with applying purge gas: firstly, it reduces environmental pollution which significantly affects climate and environment, secondly, methanol production raises and thirdly, ammonia is produced. In this study, the optimum value of recycle ratio has been achieved provided that Stoichiometry Number (SN) remains in the range of 2–2.1. SN quantifies the relationship between reactants and products in a chemical reaction. It is defined as [15]:

$$SN = \frac{\text{mol}(\text{H}_2) - \text{mol}(\text{CO})}{\text{mol}(\text{CO}) + \text{mol}(\text{CO}_2)} \quad (3)$$

If the reformed gas becomes sub-stoichiometric ($SN < 2.0$), the loop pressure decreases and methanol production falls rapidly. On the other hand, if the stoichiometry number is too high, there is a beneficial effect on by-product formation but it has a negative impact on the reformed gas consumption per ton of methanol.

2. Process description

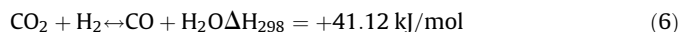
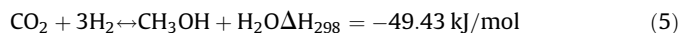
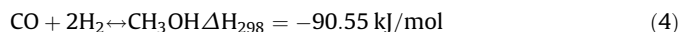
All configurations contain shell and tube heat exchanger in which the first reactor (water-cooled reactor) is combined serially with the second one (gas-cooled reactor). Cold feed synthesis gas is preheated inside the tubes of the second reactor (gas-cooled reactor), and subsequently fed into the tube side of the water-cooled reactor where the chemical reaction is initiated by catalysts. of Methanol synthesis reactions are performed over commercial $\text{CuO}/\text{ZnO}/\text{Al}_2\text{O}_3$ catalyst. In this stage, synthesis gas is partly converted to hydrocarbons. The reacted gas leaving the water-cooled reactor is directed into the shell side of the gas-cooled reactor in counter-current mode with synthesis gas flowing through the tube side. The reaction is completed in the second reactor. The next stage is purification of produced methanol. The outlet stream of reactor is cooled until that methanol is condensed. Purge gas and methanol condensate are separated in the flash drum. In commercial processes, the unconverted syngas is recycled back to the methanol converter for enhancing the overall conversion, thereby improving the process economics.

Purge gas stream is recycled to the feed of gas-cooled reactor in IC and ORC. The recycle ratio of IC is 2.19 and it is 0.41 for ORC. 7% of purge gas in IC burns in the flare gas while 59% purifies in the ORC. In ORC, a Pd–Ag membrane is proposed for hydrogen removal from the flare gas and a vanadium membrane is proffered for nitrogen removal and also a vanadium membrane layer is assisted in a reactor for ammonia production. However, the purge gas in FFC is burnt in a flare completely. A schematic diagram of ORC and IC are shown in Fig. 2(a) and (b), respectively. Moreover, a schematic diagram of FFC is depicted in Fig. 2(c).

3. Reaction scheme and kinetics

Methanol and its derivatives play a pivotal and crucial role in a wide range of industrial processes. The importance of methanol in different applications such as fuel, pesticide, and drug productions is undeniable [16]. Methanol synthesis includes three overall

reactions as follows:



Reactions (4)–(6) are not independent so that one is a linear combination of the other ones. In the present work, the rate expressions have been selected from Graaf et al. [17]. The rate equations combining with the equilibrium rate constants provide valuable source of information about kinetics of methanol synthesis [18]. The corresponding rate expressions according to the hydrogenation of CO, CO_2 and reversed water–gas shift reactions over commercial $\text{CuO}/\text{ZnO}/\text{Al}_2\text{O}_3$ catalysts are:

$$r_1 = \frac{k_1 K_{\text{CO}} [f_{\text{CO}} f_{\text{H}_2}^{3/2} - f_{\text{CH}_3\text{OH}} / f_{\text{H}_2}^{1/2} K_{P1}]}{(1 + K_{\text{CO}} f_{\text{CO}} + K_{\text{CO}_2} f_{\text{CO}_2}) [f_{\text{H}_2}^{1/2} + (K_{\text{H}_2\text{O}} / K_{\text{H}_2}^{1/2}) f_{\text{H}_2\text{O}}]} \quad (7)$$

$$r_2 = \frac{k_2 K_{\text{CO}_2} [f_{\text{CO}_2} f_{\text{H}_2}^{3/2} - f_{\text{CH}_3\text{OH}} f_{\text{H}_2\text{O}} / f_{\text{H}_2}^{3/2} K_{P2}]}{(1 + K_{\text{CO}} f_{\text{CO}} + K_{\text{CO}_2} f_{\text{CO}_2}) [f_{\text{H}_2}^{1/2} + (K_{\text{H}_2\text{O}} / K_{\text{H}_2}^{1/2}) f_{\text{H}_2\text{O}}]} \quad (8)$$

$$r_3 = \frac{k_3 K_{\text{CO}_2} [f_{\text{CO}_2} f_{\text{H}_2} - f_{\text{H}_2\text{O}} f_{\text{CO}} / K_{P3}]}{(1 + K_{\text{CO}} f_{\text{CO}} + K_{\text{CO}_2} f_{\text{CO}_2}) [f_{\text{H}_2}^{1/2} + (K_{\text{H}_2\text{O}} / K_{\text{H}_2}^{1/2}) f_{\text{H}_2\text{O}}]} \quad (9)$$

The reaction rate, adsorption equilibrium and reaction equilibrium constants which occur in the formulation of kinetic expressions are tabulated in Table 1, respectively.

Ammonia formation is a reversible exothermic reaction.



To calculate the rate of reaction, the modified Temkin equation offered by Dyson and Simon [19] in 1968 has been used.

$$R_{\text{NH}_3} = 2k \left[K_a^2 a_{\text{N}_2} \left(\frac{a_{\text{H}_2}^3}{a_{\text{NH}_3}^2} \right)^\alpha - \left(\frac{a_{\text{NH}_3}^2}{a_{\text{H}_2}^3} \right)^{1-\alpha} \right] \quad (11)$$

Table 1

The reaction rate constants, the adsorption equilibrium constants, and the reaction equilibrium constants for methanol synthesis.

	A	B
$K = A \exp(B/RT)$		
k_1	$(4.89 \pm 0.029) \times 10^7$	$-63,000 \pm 300$
k_2	$(1.09 \pm 0.07) \times 10^5$	$-87,500 \pm 300$
k_3	$(9.64 \pm 7.30) \times 10^6$	$-152,900 \pm 6800$
$K = A \exp(B/RT)$		
K_{CO}	$(2.16 \pm 0.44) \times 10^{-5}$	$46,800 \pm 800$
K_{CO_2}	$(7.05 \pm 1.39) \times 10^{-7}$	$61,700 \pm 800$
$(K_{\text{H}_2\text{O}}/K_{\text{H}_2}^{1/2})$	$(6.37 \pm 2.88) \times 10^{-9}$	$84,000 \pm 1400$
$K_P = 10^{(A/T-B)}$		
K_{P1}	5139	12.621
K_{P2}	3066	10.592
K_{P3}	−2073	−2.029

where, α is a constant which takes a value of 0.5–0.75 in literature [19].

k : Rate constant for the reverse reaction k

$$= 8.849 \times 10^{14} \exp\left(-\frac{40765 \frac{\text{kcal}}{\text{kmol}}}{RT}\right) \quad (12)$$

K_a : Equilibrium constant $\log K_a$

$$= -2.691122 \log T - 5051925 \times 10^{-5} T + 1.848863 \times 10^{-7} T^2 + \frac{2001.6}{T} + 2.689 \quad (13)$$

$$a_i : \text{Activity} \quad a_i = y_i \phi_i P \quad (14)$$

The applied equations are the experimental ones for fugacity coefficient of hydrogen, nitrogen and ammonia [19].

$$\phi_{H_2} = \exp\left(\frac{-3.84027^{0.125} + 0.541}{P} - e^{(-0.12637^{0.5} - 15.98)} P^2 + 300[e^{(-0.011901^T - 5.941)}](e^{\frac{P}{300}})\right) \quad (15)$$

$$\phi_{N_2} = 0.93431737 + 0.2028538 \times 10^{-3} T + 0.295896 \times 10^{-3} P - 0.270727 \times 10^{-6} T^2 + 0.4775207 \times 10^{-6} P^2 \quad (16)$$

$$\phi_{NH_3} = 0.1438996 + 0.2028538 \times 10^{-2} T - 0.4487672 \times 10^{-3} P - 0.1142945 \times 10^{-5} T^2 + 0.2761216 \times 10^{-6} P^2$$

4. Mathematical model

4.1. Fixed bed reactors

A one dimensional heterogeneous model, which is a conventional model for a catalytic reactor with heat and mass transfer resistances, has been developed for this reactor in order to determine the concentration and temperature distributions inside the reactor. The reactor is simulated by MATLAB software. In this model the following assumptions are used:

The gas mixture is an ideal gas in both catalytic reactors.

Both of the reactors are operated at steady-state conditions.

Radial variations in both beds are negligible (one-dimensional model).

Axial diffusion of mass and heat are negligible.

Bed porosity in axial and radial directions is constant.

Laminar plug flow is employed in both reactors.

It is assumed that the chemical reactions take place only in the catalyst particles.

Heat loss to surrounding is neglected.

The mathematical model of the fixed bed reactor is not so complicated and it has been presented a lot in the literature.

4.1.1. Solid phase

The mass and energy balances for the solid phase are as follows:

$$a_v c_j k_{gi} (y_i^g - y_i^s) + \eta r_i \rho_b = 0 \quad (18)$$

$$a_v h_f (T^g - T^s) + \rho_b \sum_{i=1}^N \eta r_i (-\Delta H_{f,i}) = 0 \quad (19)$$

where y_i^s and T^s are the solid-phase mole fraction and temperature for components (5 components) i in the reactor, respectively and η is the effectiveness factor.

4.1.2. Fluid phase

The following mass and energy balance equations are developed for the fluid phase:

$$-\frac{F_j dy_j^g}{A_c dz} + a_v c_j k_{gi} (y_i^s - y_i^g) = 0 \quad (20)$$

$$-\frac{F_j C_{pj}^g dT_j^g}{A_c dz} + a_v h_f (T^s - T^g) \pm \frac{\pi D_i U_{1-2} (T^{\text{shell}} - T_1^{\text{tube}}) = 0 \quad (21)$$

where y_i^g and T^g are the fluid phase mole fraction and temperature for component i in the reactor, respectively. The energy equation consists of the heat transfer by convection and the heat exchanged between the fluid phase and solid phase. The boundary conditions are as follows:

$$z = 0, y_i^g = y_{i0}^g, T^g = T_0^g, P^g = P_0^g \quad (22)$$

where y_{i0}^g, T_0^g and P_0^g are the fluid phase mole fraction, reactor inlet temperature, and pressure at the entrance of the reactor, respectively.

4.2. Purification unit

A one dimensional model is considered to the feed side of vanadium membrane as a purification unit.

$$-\frac{F dy_i}{A_c dz} - \frac{P_M}{\delta} (\sqrt{P_h x_j} - \sqrt{P_l y_j}) \times \pi \times D_m \times L_m = 0 \quad (23)$$

where L_m and D_m are the length and diameter of membrane, respectively. x_j is the mole fraction of component j in the tubular membrane and y_j is the mole fraction of component j in the permeable side of the tubular membrane.

Furthermore, permeable side of vanadium membrane is modeled as a one dimensional model. Mass balance equation for ammonia is:

$$\frac{2F_{N_2} dy_i}{A_p dz} = \eta r_{NH_3} \quad (24)$$

Mass balance equation for the other components is:

$$\frac{F_i dy_i}{A_p dz} = \frac{P_M}{\delta} (\sqrt{P_h x_j} - \sqrt{P_l y_j}) \times \pi \times D_m \times L_m \quad (25)$$

A_p is the area of permeable side. The energy balance is ignored in the mathematical modeling of purification unit.

Table 2

Comparison between model results with plant data for fresh catalyst.

Product condition	Plant	Predicted	Error %
Composition (mol%)			
CH ₃ OH	0.104	0.1023	−3.4
CO	0.0251	0.0228	−4.38
CO ₂	0.0709	0.0764	−9.16
H ₂ O	0.0234	0.0211	−9.82
H ₂	0.5519	0.5323	−3.55
N ₂ /Ar	0.0968	0.0905	−6.5
CH ₄	0.114	0.103	−9.64
Temperature (K)	495	489.5	−1.2
SN	3.2	3.4	6.25

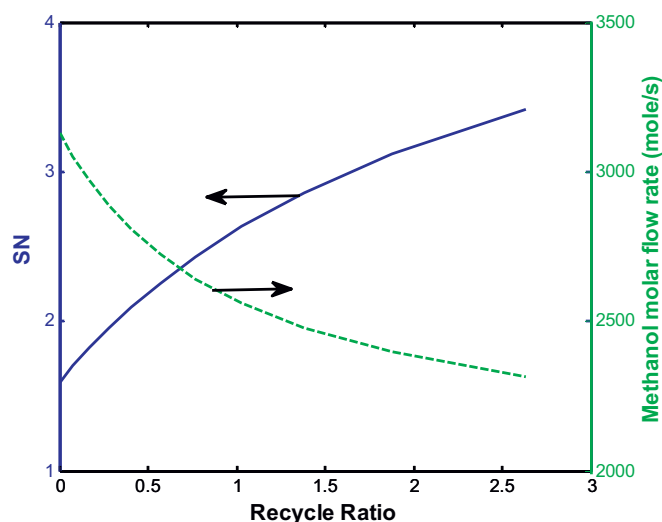


Fig. 3. SN and methanol molar flow rate versus the recycle ratio in ORC.

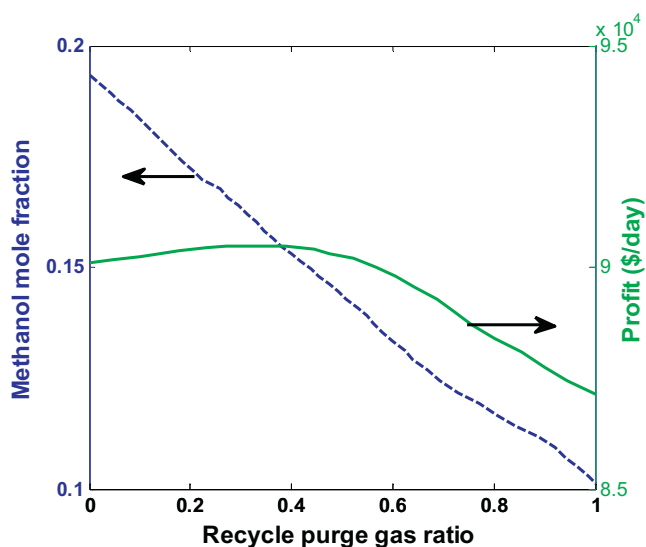


Fig. 4. Methanol mole fraction and profit versus the recycle purge gas ratio.

5. Optimization

In the simplest case, an optimization problem consists of maximizing or minimizing a real function by systematically choosing input values from within an allowed set and computing the value of the function. The generalization of optimization theory and techniques to other formulations comprises a large area of applied mathematics. It is considerable to note that optimization plays a vital role in industrial operations, manufacturing and engineering activities in order to be more efficient. A mathematical model, which predicts process behaviors, is required to optimize chemical and related processes [20].

5.1. Differential Evolution (DE) method

Differential Evolution (DE) method, a population based algorithm like genetic algorithms, is a simple structure and significantly fast method in numerical optimization [21]. According to the type of problem for which DE should be applied, various strategies could be adapted in DE algorithm. The choice of DE parameters including Differential Weight (F), Crossover Probability

Constant (CR) and Number of Population (NP) can have a significant impact on optimization performance.

Selecting the DE parameters that yield good performance has therefore been the subject of a wide range of researches [22,23]. “DE/best/1/bin” with NP=100, CR=0.8 and F=1 are strategy and parameters utilized in the current optimization. DE method has been applied in numerous processes to optimize the operating conditions [24].

5.2. Objective function and constraints

In the current study, maximizing methanol production is considered in the objective function defined as follows:

$$F = y_{\text{CH}_3\text{OH}}(\text{Methanol mole fraction}) \quad (26)$$

One decision variable, the recycle ratio (Φ) is measured during the optimization process. The recycle ratio has been selected because of its changes in the feed of IC, which directly affects the feed of reactors. On the other hand, increasing the recycle ratio leads to makeup gas decreasing (fresh feed) which highly influences the rate of methanol production. Therefore, the feed function related to the fraction of recycling purge gas is:

$$\begin{aligned} (\text{Feed})_{\text{new}} &= \text{MUG} + \text{purge gas recycled} \\ &= \text{MUG} + \Phi \times \text{MUG} = \text{MUG} \times (1 + \Phi) \end{aligned} \quad (27)$$

The range of decision variable (Φ) is:

$$\Phi > 0 \quad (28)$$

The recycle ratio varies from 0 to ∞ , when the recycle ratio is zero the reactor feed is a fresh feed. In contrast, the entire reactor feed is purge gas when the recycle ratio is infinity. As a consequence, Eq. (28) is defined as a constraint. Additionally, the most important parameter in methanol synthesis is SN which is needed slightly above 2 for this purpose. In this study, SN changes between 2–2.1 as a constraint.

$$2 < \text{SN} < 2.1 \quad (29)$$

Constraints are considered during the optimization process. A penalty function method has been used in order to eliminate automatically the unacceptable results. In this case, 10^5 is selected as a penalty parameter, however its value depends on the order of magnitude of the decision variable. The objective function that should be minimized is:

$$\text{OF} = -(y_{\text{CH}_3\text{OH}}) + 10^5 \times (G_1 + G_2 + G_3) \quad (30)$$

$$G_1 = \max(0, (0 - \Phi)) \quad (31)$$

$$G_2 = \max(0, (2 - \text{SN})) \quad (32)$$

$$G_3 = \max(0, (\text{SN} - 2.1)) \quad (33)$$

A set of above mentioned differential-algebraic equations (DAE) has been developed for the modeling of methanol reactors. The energy and mass balance equations are coupled with non-linear algebraic equations of kinetic model, transport properties and other auxiliary correlations. A backward finite difference approximation is applied to solve these equations.

6. Model validation

The steady state model validation is performed between the pilot plant data reported by domestic petrochemical company and

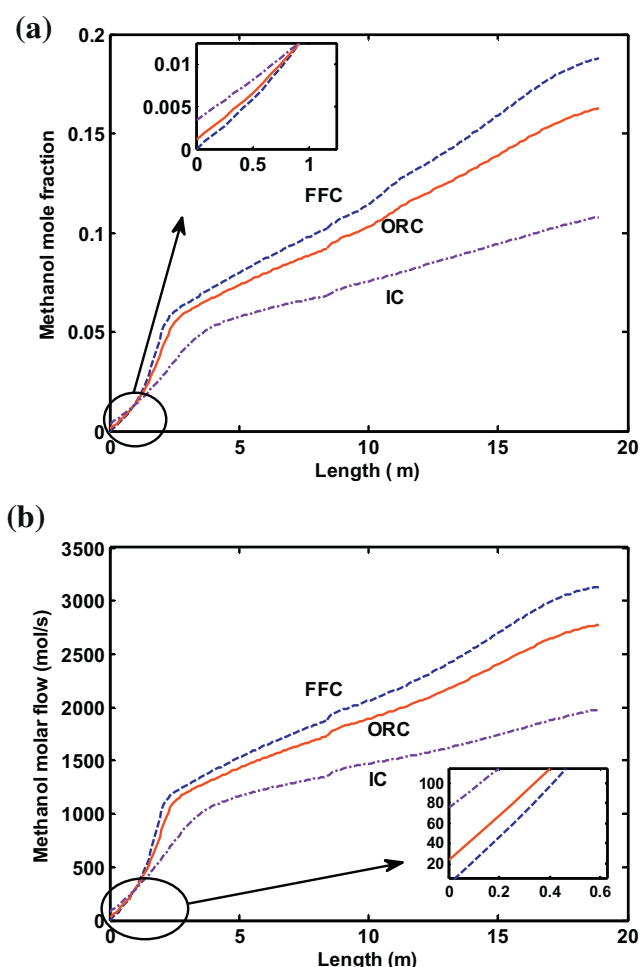


Fig. 5. The comparison between the (a) methanol mole fraction (b) molar flow rate of methanol in IC, FFC and ORC.

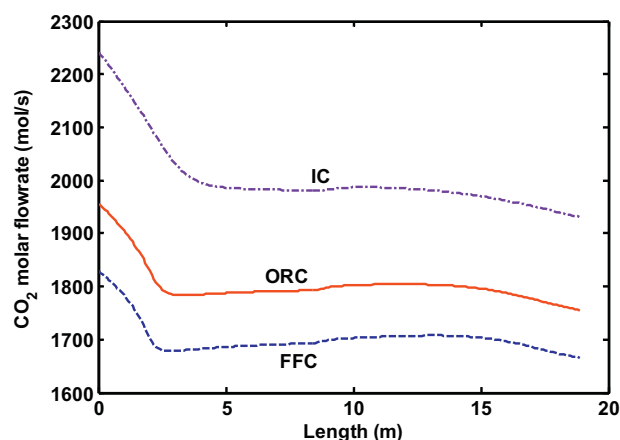


Fig. 6. Carbon dioxide molar flow rates along IC, FFC and ORC.

7. Result

In this study, the percentage of recycle stream is optimized via DE method in order to maximize the methanol mole fraction. The effect of the variation of recycle ratio on stoichiometry number (SN) and methanol molar flow rate in ORC is depicted in Fig. 3. As seen, by decreasing the stoichiometry number, the rate of methanol production increases. The maximum desired value of SN is 2.1 and there is the best ratio of hydrogen to carbon monoxide in this range of SN; therefore, the rate of methanol production increases. The stoichiometry number is 3.4 in IC in which the ratio of hydrogen to carbon monoxide is in a high level. So, the methanol production reduces and also a large amount of unreacted hydrogen will remain in the output.

In this study, the equation for profit, which has been obtained from methanol production unit, is considered as follows:

$$F = (\text{produced methanol} \times \text{price} - \text{Fresh Feed} \times \text{price}) \quad (34)$$

$$F = (\text{Molar flow rate of methanol} \times \text{molecular weight})(\text{kg/hr})$$

$$/1000 \times 400 (\$/\text{ton}) - \text{feed}(\text{m}^3/\text{h}) \times 0.25 (\$/\text{m}^3)$$

The effect of the variation of recycle purge gas ratio on profit and methanol mole fraction has been illustrated in Fig. 4. As seen, methanol production decreases by increasing the recycle purge gas ratio, whereas the profit has the maximum value in the recycle purge gas ratio of 0.41. As the recycle purge gas ratio rises, the amount of inert gases increases gradually and therefore the rate of methanol production decreases. By increasing the recycle purge gas ratio despite less amount of fresh gas is required, the rate of methanol production reduces. At first, by increasing the recycle purge gas ratio the unit profit increases to its highest value of 0.41 due to the fresh gas reduction in the feed and after that the unit profit decreases because of a reduction in the rate of methanol production. In fact, after the recycle ratio of 0.41, the production declining has significant impact on the unit profit compared to the feed reduction.

The result comparisons of Optimized Recycle Configuration (ORC), Fresh Feed Configuration (FFC) and Industrial Configuration (IC) are investigated through Figs. 5–10. As it is clearly known, there is no recycle stream in FFC and IC is related to a configuration which has 2.19 recycle ratio (it is 95% purge gas stream).

A comparison between the methanol mole fraction in IC, FFC and ORC has been portrayed in Fig. 5(a). The mole fraction of methanol, as a main product, increases considerably in FFC in comparison with IC. There is a remarkable increase in the methanol mole fraction in FFC which is attributable to lack of impurity (nitrogen and methane) in this configuration. As shown in Fig. 5(a), at the entrance of the reactor, the mole fraction of methanol is different for three mentioned configurations. There is no methanol in the feed of FCC, whereas methanol mole fraction in the feed of ORC is approximately 0.001 which is because of a proportional amount of residual methanol in the form of gas after passing a methanol separator. Therefore, methanol is added to the feed of ORC and IC with the recycling gas stream. The inlet mole fraction of methanol in IC is slightly more than the one in the other configurations.

Fig. 5(b) gives a wealth of information about methanol molar flow rate. The highest methanol molar flow rate is related to FFC; furthermore, the molar ratio of FFC to ORC is approximately 1.2 in methanol molar flow rate. Although the molar flow rate of methanol in FFC is almost more than ORC, the Stoichiometry Number (SN) in FFC is more than 2.1, which is an undesired result. In ORC, methanol molar flow rate increases to almost 41% more

the mathematical modeling of IC. The results of IC model and the corresponding observed data of the pilot plant are presented in Table 2. A good agreement is observed between the modeling results and the pilot plant data. Therefore, this mathematical model performs well under the industrial conditions.

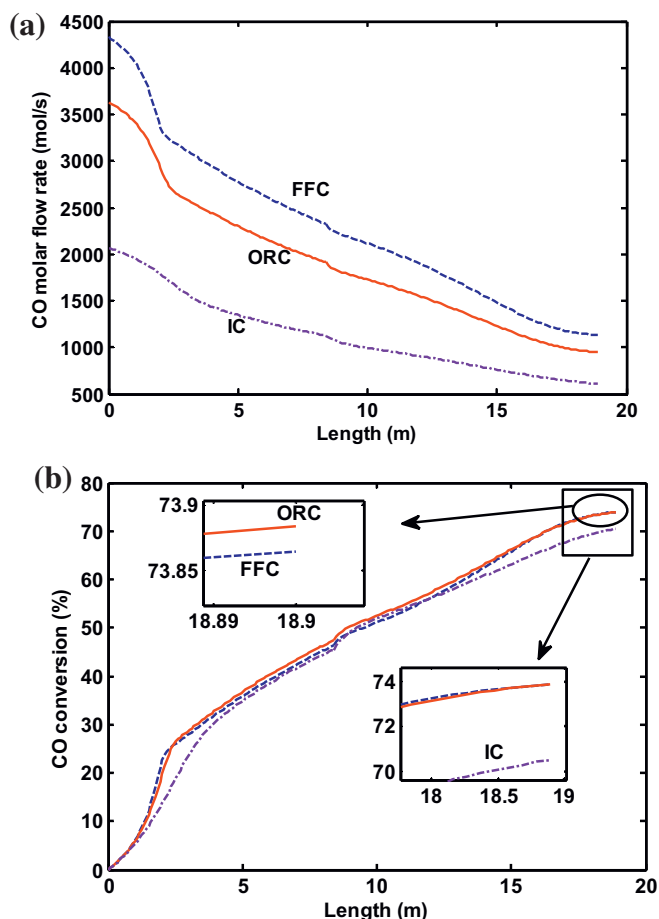


Fig. 7. (a) The molar flow rate of carbon monoxide (b) The carbon monoxide conversions along IC, FFC and ORC.

than IC. Consequently, the profit obtained from ORC is much more than IC.

The reason of lower production of methanol in ORC in comparison with FFC is the inert gas increase in reaction medium. On the other hand, the cause of methanol production increase in ORC compared to IC is a decrease in the rate of recycle gas stream. However, by increasing the rate of recycle gas stream, less amount of gas is burnt in the flare and it is environmentally friendly, it is not economical due to reduction of production and profit rates.

Carbon dioxide molar flow rate as a function of reactor length has been demonstrated in Fig. 6 for IC, FFC and ORC. There is the highest value of inlet carbon dioxide mole fraction in IC in comparison with the one in FFC and ORC. Besides, the molar flow rate of carbon dioxide in ORC is much more than FFC. In general, the inlet carbon dioxide mole fraction is higher than the outlet carbon monoxide mole fraction, which is due that carbon dioxide is converted to methanol in different ways. Consequently, as the percentage of recycle flow rate increases, there is more carbon dioxide in the feed of reactor; however the rate of carbon dioxide has no significant effect on methanol production. The key reason of low methanol production via increasing the amount of carbon dioxide in the recycle stream is that the reaction rate of methanol production via carbon monoxide is higher than the one by carbon dioxide. Although carbon dioxide is added to the feed of reactor by recycling more purge gas which leads to reduction in methanol production, the process is an environmentally friendly process. As shown in Fig. 6, the tendency of carbon dioxide consumption is high in the first two meters of the reactor, which is because of methanol second reaction. Then the consumption of carbon

dioxide remains constant until the end of the first reactor. The second reaction of methanol production is a reversible reaction which reaches equilibrium in the first two meters of the first reactor and then the reaction carries out slowly in the rest of the reactor length. As seen in Fig. 6, there is a negligible enhancement of carbon dioxide molar flow rate at the entrance of the second reactor, which is owing to the endothermic water-gas shift (WGS) reaction. In the first reactor, the outlet flow temperature is high, which can be used for the second reactor as an endothermic WGS reaction. In WGS reaction carbon dioxide is produced from carbon monoxide and water. The second reactor is a shell and tube reactor; furthermore, syngas flows in the reactor tubes as a coolant. Meanwhile a slight amount of endothermic WGS reaction takes place at the entrance of the reactor; thus, the amount of carbon dioxide production is negligible. When the reaction media is cooled with syngas, a large amount of exothermic reaction of methanol production is carried out and carbon dioxide is consumed.

The molar flow rate of carbon monoxide versus reactor length has been shown in Fig. 7(a) for three different configurations. The lowest and highest amount of carbon monoxide happens at the entrance of IC and FFC, respectively. The lowest amount of carbon monoxide at the entrance of IC is related to the recycling 93% of purge gas stream, which mostly includes carbon dioxide and hydrogen. As a considerable amount of carbon monoxide is consumed in the input stream, the unconverted carbon monoxide is negligible in the outlet. The inlet carbon monoxide in ORC is less

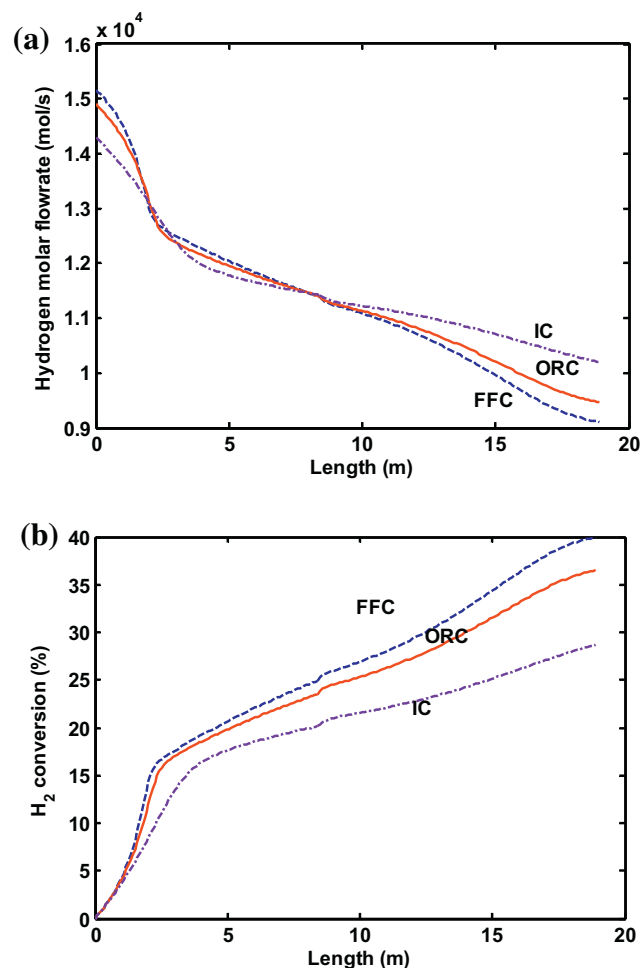


Fig. 8. (a) The molar flow rate of hydrogen (b) Hydrogen conversion along IC, FFC and ORC.

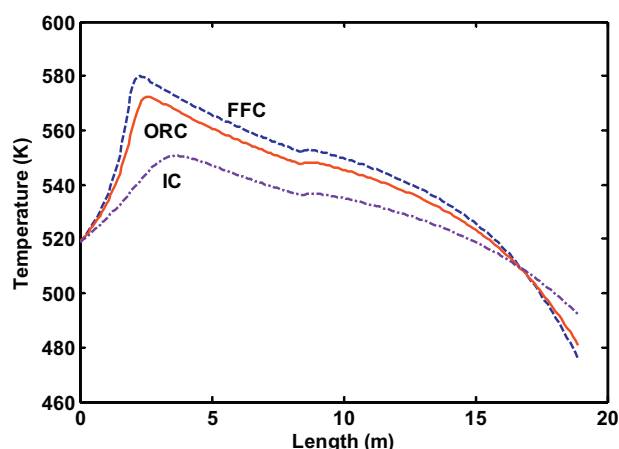


Fig. 9. Comparison between temperature of IC, FFC and ORC.

the reaction takes place so quickly. Therefore, the temperature

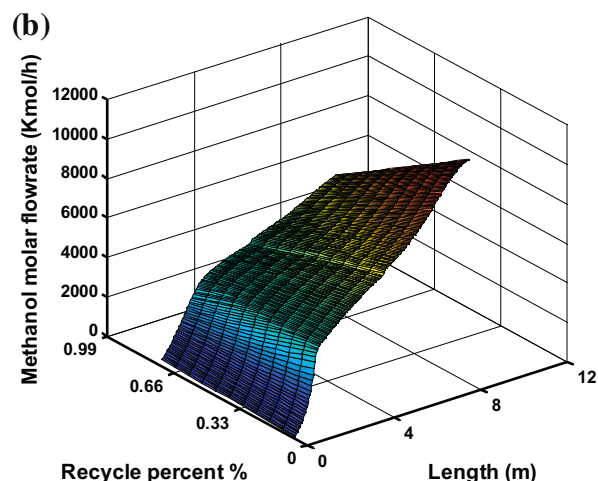
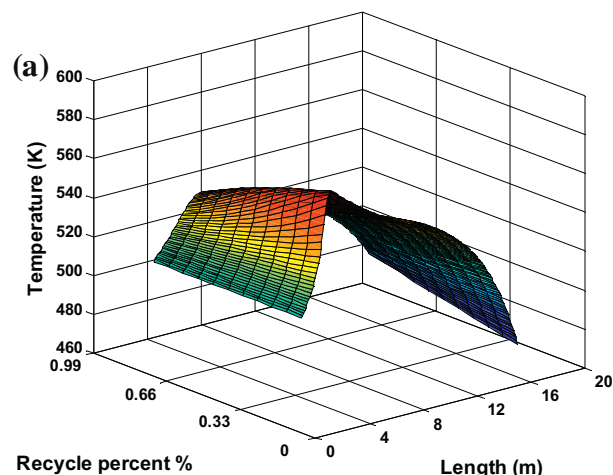


Fig. 11. The effect of recycle flow rate to the reactor (a) on temperature (b) on the rate of methanol production.

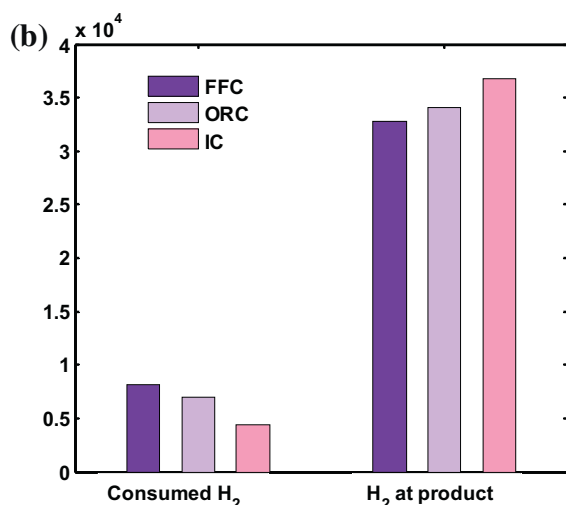
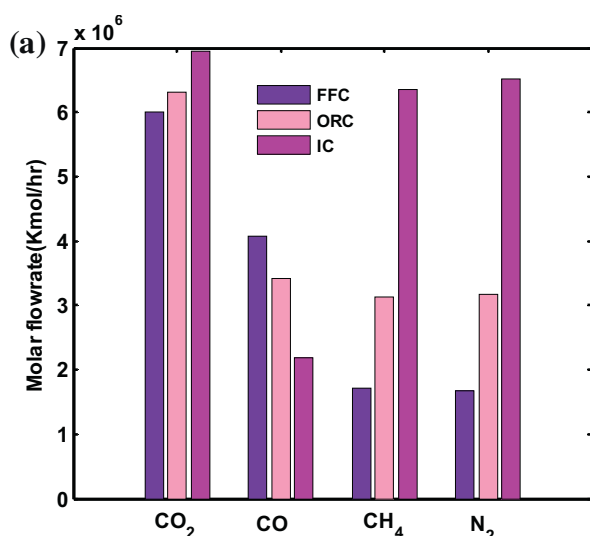


Fig. 10. Comparison between (a) the molar flow rates of CO_2 , CO , CH_4 and N_2 (b) the consumed hydrogen and hydrogen in the product.

increases and the mole fraction of carbon monoxide reduces abruptly. As the temperature rises and it reaches equilibrium, the reaction carries out slowly and the mole fraction of carbon monoxide decreases along the reactor length. The conversion percent of carbon monoxide as a function of reactor length has been depicted in Fig. 7(b) for IC, FFC and ORC. As clearly seen, the conversion percent of hydrogen and carbon monoxide increase in all configurations owing to carrying the reactions. There is a less amount of carbon monoxide conversion in IC than the other two configurations. In addition, the highest conversion is referred to ORC. There is approximately 0.02% difference between conversion of ORC and FFC which is negligible, whereas the difference between the conversion of ORC and IC in the outlet is about 3.46%.

Hydrogen molar flow rate versus reactor length has been illustrated in Fig. 8(a) for IC, FFC and ORC. As obviously shown, maximum and minimum molar flow rate of hydrogen is related to FFC and IC, respectively. Hydrogen is consumed very fast at the entrance of the first reactor. This is attributed to the high reaction rate which declines gradually along the reactor length. As the rate of methanol production rises, a sizeable amount of hydrogen is consumed. Ultimately, maximum molar flow rate of hydrogen is obtained in the outlet of IC. Because the stoichiometric ratio has not been observed, the rate of hydrogen is much more than the rate

of carbon monoxide in IC; thus, a high rate of hydrogen will remain in the outlet. Approximately 7% of the purge gas is burnt in a flare in the case of IC. About 60% of this purge gas is valuable hydrogen. So in this case, in addition to decline in production and profit rates, a considerable amount of hydrogen, which has been produced with a high cost in the reformer unit, is wasted.

Hydrogen conversion percent as a function of reactor length is presented in Fig. 8(b). Hydrogen conversion increases dramatically in the first two meters of the reactor and afterward it rises gradually. It is evident that hydrogen conversion reaches its highest value in FFC which leads to a reduction in hydrogen molar flow rate in the outlet of reactor. Since the amount of SN reaches the lowest level, the ratio of hydrogen to carbon monoxide will remain in a desirable range of 1.8, therefore, the maximum hydrogen conversion has been observed in this case. Furthermore, at the end of the reactor, the lowest value of hydrogen conversion is related to IC.

Fig. 9 provides a great deal of information about temperature along the reactor length of three different configurations. However the feed temperature is exactly equal for three configurations, the IC temperature is the least one along the first reactor. This occurs because of the difference between the reaction rates and also the increasing in feed impurities. According to the graph lines, input temperature for the second reactor is different in IC, ORC and FFC. Although at the entrance of the second reactor there is the least temperature in IC, the least temperature in the outlet of the reactor is related to FFC and ORC owing to temperature drop.

Molar flow rate of carbon dioxide, carbon monoxide, methane and nitrogen at the outlet of three configurations has been illustrated in Fig. 10(a). Minimum and maximum value for molar flow rate of carbon dioxide, methane and nitrogen is related to FFC and IC, respectively. As obvious, the molar flow rate of nitrogen and methane is rising rapidly in IC. In order to increase the product in IC, it could be proposed that a separation unit is placed in the route of recycle stream for nitrogen and methane removal and they should be replaced them by fresh gas at each step.

Fig. 10(b) demonstrates the rate of consumed hydrogen in comparison with the amount of hydrogen in product in IC, FFC and ORC. As seen, the highest and least amount of unconsumed hydrogen is related to FFC and IC, respectively.

Fig. 11(a) gives information about the influence of flow rate percent which is recycled to the reactor on temperature along the reactor length. Provided that more purge gas is recycled, the temperature would have less jumping at the entrance of the reactor; however, the temperature will be great at the end of the second reactor. The impact of recycle flow rate to the reactor on the rate of methanol production along the reactor length is depicted in Fig. 11(b). As purge gas recycling increases, the methanol production reduces.

8. Conclusion

Generally the purge gas of different processes contains a lot of unconverted components which can be reused. There is two advantages in recycling purge gas stream: firstly, it reduces environmental pollution which considerably affects climate and environment. Secondly, the unit profit increases. The purge gas stream is usually sent and burnt in the flare which strongly damages the environment. Furthermore, there are valuable components in such gas stream which can be potentially converted to the valuable products. The optimized value of recycle ratio of purge gas to the feed of methanol production has been investigated in this study. The optimum recycle ratio is approximately 0.41 (42.5% of purge gas stream is recycled) in order to reach the maximum methanol mole fraction. However, the recycle ratio is approximately 2.19 (93% of purge gas stream is recycled) in

methanol production unit in the domestic petrochemical industry. As the recycle ratio is reduced to 0.41, SN will be decreased to 2.1. Stoichiometric number represents the ratio of hydrogen to carbon monoxide in the feed of methanol synthesis. The optimal range of SN is from 1.8 to 2.1 and methanol production is in the highest level in this range. Because 93% of purge gas stream is recycled to the methanol synthesis reactors in IC, the stoichiometric number is out of the optimal range and the rate of methanol production is reduced. So, there is almost 13% increase in the rate of hydrogen in the outlet of IC compared to FFC. The rate of methanol production in ORC is almost 1.17 more than the one in IC. The rate of methanol production increases with decreasing the recycle ratio and increasing nitrogen separation in the recycled gas stream, because nitrogen is considered as an impurity component in ORC. Two separation units are proposed after methanol synthesis reactors in ORC in the current study. One of them separates nitrogen and recycles the purge gas without nitrogen to the fresh feed of methanol synthesis. However, the other one separates hydrogen from the flare gas stream. Hydrogen is utilized as a sweep gas for nitrogen separation by a novel metallic membrane. The purification units include a vanadium membrane which separates nitrogen from the purge gas; furthermore, a Pd–Ag membrane is applied in order to separate hydrogen from the flare gas. Consequently, ammonia is produced via nitrogen reaction plus hydrogen along the vanadium membrane. Accordingly, the remaining hydrogen in the purge gas, which should be burnt in the flare, is optimized and used in the production of ammonia in ORC. But hydrogen in the 7% of purge gas in IC and 100% of purge gas in FFC are burnt, while it has been produced with a high cost in the reformer unit. The proposed configuration of ORC causes a profit increase compared to IC and also it is a much more environmentally friendly configuration in comparison with FFC.

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